# Resilient decentralized communication for unstructured peer-to-peer networks

## From Sparsity to Reliability: the Decentralized Path to Free Speech

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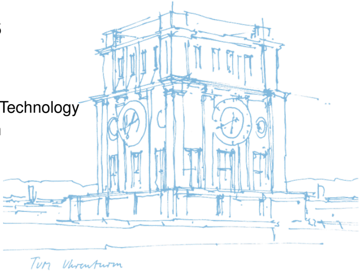

Provide resilient distributed communication during Internet shutdowns in case of disasters

# Introduction

## Problems

- Consolidation of power in Internet infrastructure due to running services, protocol, and algorithms in a few closed providers
- Centralization enables both rapid information diffusion but also censorship, failure points, and control, worsened by disasters and political unrest


# Introduction

## Solution Approach

To solve fragile consolidation of power, resilient decentralization is key!

- Providing connectivity without infrastructure: p2p networks
  - Structured: Kademlia, BitTorrent, more recently DLT: ETH, Bitcoin, Ripple XRP, etc
  - **Unstructured**: FireChat, Twimight, Briar, bitchat, Amigo

To enable sustainable critical communication for all: use efficient ubiquitous p2p protocols

- **BLE**
- Others: WiFi-Direct, Zigbee, LoRA and others

**Research Question** How to provide resilient decentralized communication over unstructured p2p networks using efficient p2p communication protocols in the presence of network disruptions?

**Approach** Collect empirical data to parameterize signal, simulate distributed network topologies, measure resilience w.r.t centralization metrics

# Outline

- Distributed resilience as overall similar number of links in topology
- Compare different metrics: $L_0$ norm, $\mathscr{S}$-Score, Gini

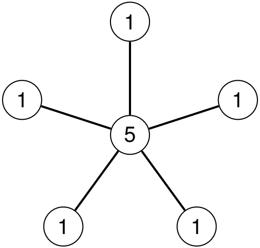


Figure 1: Star topology, $L_0 = 1.0$

# State-of-the-Art

## Resilience Metrics: the $\mathscr{S}$-Score

- Metric coming from US antitrust laws to measure market concentration
- Measures similarity to maximally centralized topology

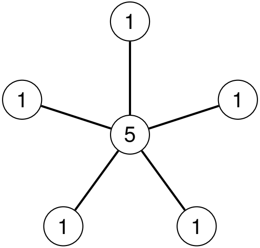


Figure 2: Star topology, $\mathscr{S} = 1.0$

# State-of-the-Art

## Resilience Metrics: Six Criteria of Sparsity

1. Robin Hood: Stealing from the rich and giving to the poor decreases sparsity
2. **Scaling**: multiplying all values by a constant does not change sparsity
3. Rising Tide: Adding a constant to all values decreases sparsity
4. Cloning: Cloning the vector does not change sparsity
5. Bill Gates: Adding a large value to the vector increases sparsity
6. Babies: Adding a zero to the vector increases sparsity

# State-of-the-Art

## Resilience Metrics: the $\mathscr{S}$-Score Sensitivity

- High sensitivity to concentration through maximally centralized topology: detect SPOF
- $\mathscr{S}$-Score does not satisfy **Scaling** axiom

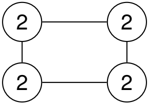

Figure 3: Topology [2, 2, 2, 2] with $\mathscr{S} = 0.75$

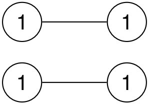

Figure 4: Topology [1,1,1,1] with $\mathscr{S} = 0.5$

## State-of-the-Art

### Resilience Metrics: the Gini Coefficient

- Popular metric used to compare wealth distributions of populations
- Measures inequality of node degrees in a topology

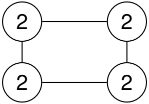


Figure 5: Topology [2, 2, 2, 2] with $G = 0$

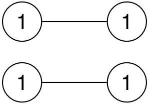


Figure 6: Topology [1,1,1,1] with $G = 0$

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# Setup

## Testbed

- **What** Measure real-life outdoor signal propagation of BLE: measure RSSI at different distances

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- **Why** TODO

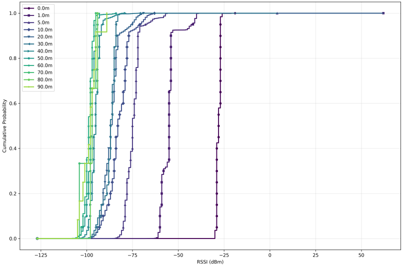


Figure 7: RSSI CDF

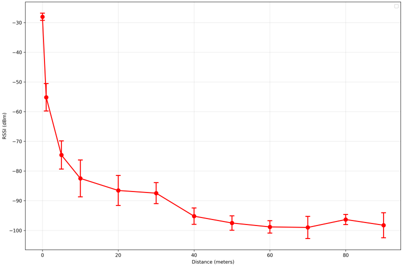


Figure 8: Mean RSSI vs. Distance

# Setup

## Testbed

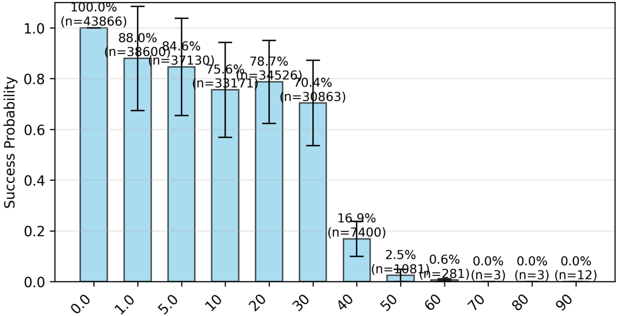


Figure 9: Advertisement Success Probability vs. Distance

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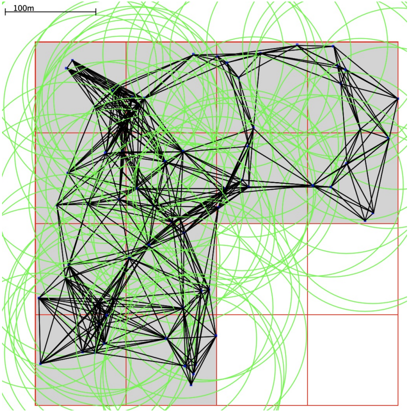


Figure 10: Simulation Area

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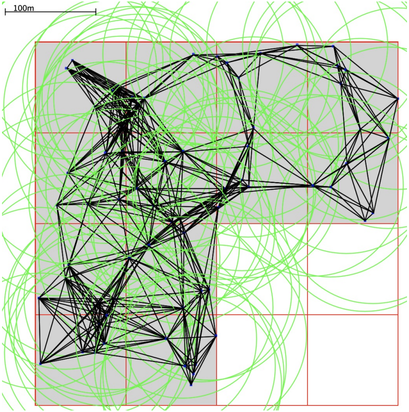


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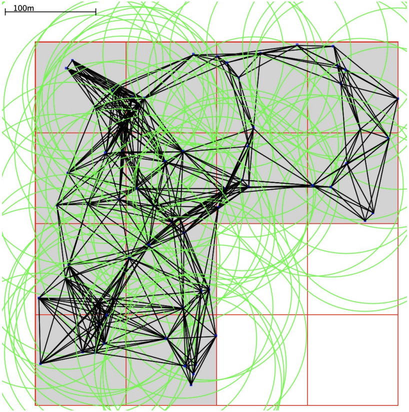


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- 12 cluster cells, 6 hosts per cell, static movement model
- Epidemic Router, same amount of messages sent between each host pair
- 25 Simulator runs consist of: network interface communication range, constrained maximum peer degree

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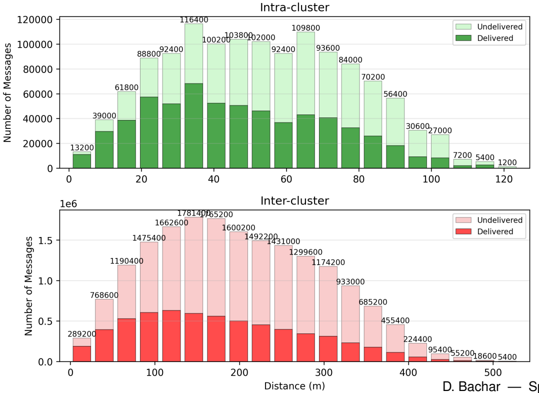

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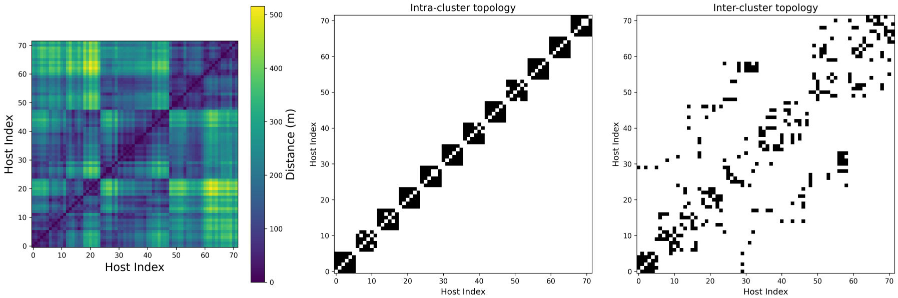


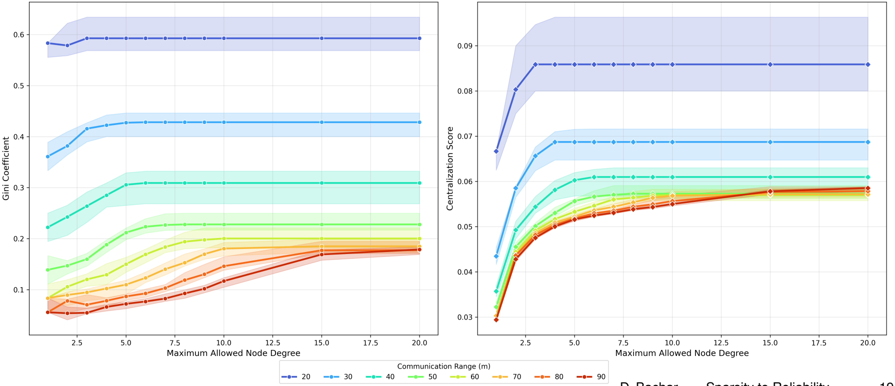

Figure 12: Topological Matrix for communication range 100m and max peer degree 5

# Outline

# Evaluation

## Model Insights

- Gini much more indicative of resilience, as $\mathscr{S} - Score$ saturates very quickly, not indicative of sparsity
- Centralization correlates with an increase in peer degree, especially in higher ranges
- Increasing maximum allowed peer degree increases centralization
- In lower communication ranges, increasing the maximum node degree has almost no effect on centralization



# Evaluation

## Model Insights

- Increasing the maximum peer degree beyond a threshold decreases latency for long-distance messages
- Short-distance messages are penalized by a high peer degree due to increased contention
- Long tail of short-distance indicates most direct links are short ranged

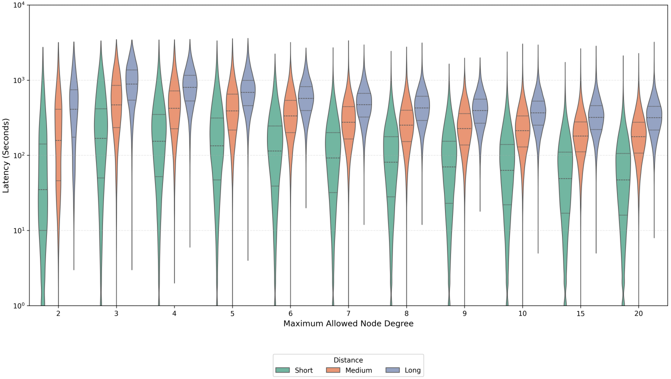



Figure 14: Latency vs. Max Peer Degree at BLE 1M PHY Range

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- Increasing centralization for short-range messages penalizes latency due to contention

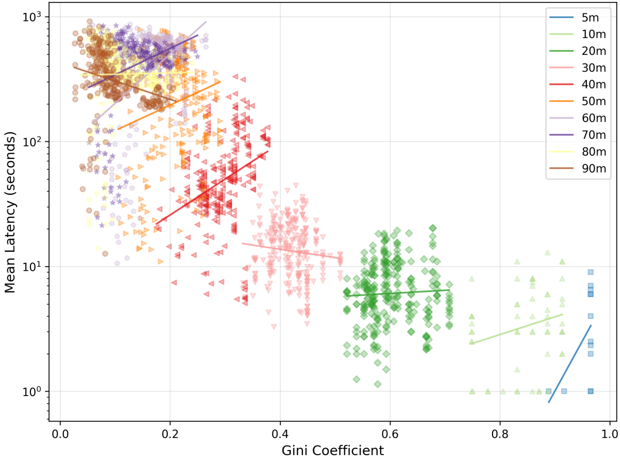


Figure 15: Latency vs. Gini Coefficient

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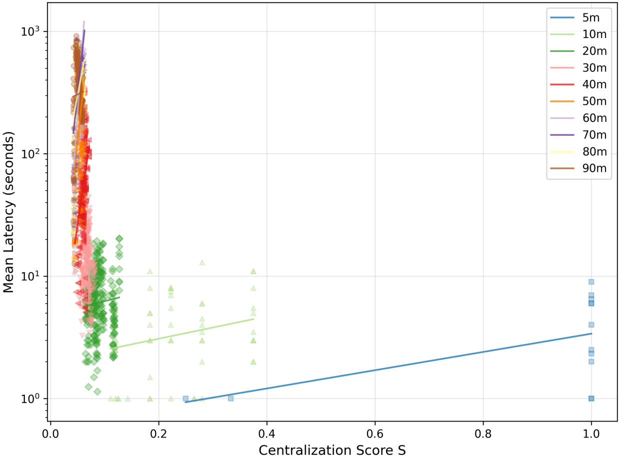


Figure 16: Latency vs. $\mathscr{S}$ — *Score*

# Outline

# Discussion

- Gini coefficient is a better indicator of resilience than the $\mathscr{S}$-Score
- $\mathscr{S}$-Score much more sensitive to extreme centralization, detecting SPOFs
- Design considerations for establishing P2P BLE networks: consider centralization and intended message range when designing P2P networks, communication-intense algorithms reduce resilience
- Gossip without consideration of peer degree can lead to communication bottlenecks and reduced resilience: consider global goal parameters such as resilience when designing p2p communication algorithms!

# Outline

# Future Work

- Explore other resilience metrics, e.g. Freedman Centralization Score
- Introduce edge churn based on advertisement success probability plot
- Measure throughput and latency in testbed based on BLE transport (unicast P2P instead of broadcast)
- Repeat empirical experiment for longer time, use the same environment for low and high ranges, compare different times of day, use custom Bluetooth stack for more control (control channel)
- Repeat experiment using mobile phones
- MA: Build p2p peering algorithm based on design considerations from this work, measure resilience in practice